Data Management Plan

CNM ? Semiconductor Device Characterization Data (180 nm CMOS)
Template: Horizon Europe DMP v1.1 (01.04.2022)
Version 1.1 | Date: 10/06/2025 | Public | Uzziel Perez
uzzielperez25@gmail.com | github.com/uzzielperez/CNM

History of changes

Version	Publication date	Change
1.0	10/06/2025	Initial DMP; FAIR inventory; EU
1.1	10/06/2025	Analysis pipelines (src/); forma

1. Data summary

1.1 Purpose of data generation and relation to project objectives

The CNM repository collects and curates semiconductor device characterization data for 180 nm CMOS technology. The data support device-level and circuit-level research objectives: Data enable comparison of measured versus simulated device behaviour, production of publication figures, and documentation of calibration conditions for reproducibility. Technology context: 180 nm PDK; nominal supply 1.8 V; process corners tt, ff, ss, fs, sf.
? Extraction of DC parameters (threshold voltage, Ion, Ioff, subthreshold swing)
? Validation of process-corner and Monte Carlo simulation setups
? Analysis of 1/f noise and AC frequency response
? Wafer-level yield and figures-of-merit benchmarking

1.2 Re-use of existing data

Asset ID	Filename	Generated / Reused	Origin	Purpose
D1	march8.csv	Generated	Lab IV measurement	DC sweep validation
D2	copy of copy of IV_curves.xlsx.c	Generated	Spreadsheet export	Raw IV reference
D3	noise_NEW2.dat	Generated	Noise analyser	1/f noise characterisation
D4	asdfgraph.png.csv	Generated	Plot export	Trace archival
D5	data1.csv	Generated	Bench measurement	Sample V?I curve
D6	Measurements - Juan - version2-	Generated	Post-processed lab data	Calibrated DC parameters
D7	untitled2.csv	Generated	Switch bench	Ron/Roff ratios
D8	FINAL_v3_USETHIS.csv	Generated	AC analyser + processing	Figure-ready frequency response
D9	definitive_DEFINITIVE.csv	Generated	Wafer test	Yield / fmax benchmarking
D10	docs/*.txt	Generated	Lab notes	Provenance and methods

No third-party datasets are currently reused. External 180 nm PDK model files are referenced in documentation but not redistributed due to foundry intellectual property restrictions (see Sections 2.2 and 6).

1.3 Types and formats of data

Data type	Format	Generation software	Open format
Tabular measurements	CSV (UTF-8, comma-separated)	Instruments, spreadsheets, Pytho	Yes
Noise spectra	DAT (ASCII, whitespace-delimited	Noise analyser export	Yes
Lab notes	TXT (UTF-8)	Plain text editor	Yes
Analysis code	PY, M, OGS	Python, MATLAB, Origin LabTalk	Yes

Preferred formats follow UK Data Service recommendations: plain text and non-proprietary tabular formats.

1.4 Provenance

Pipeline stage	Repository path	Rule
Raw instrument output	data/01_raw/	Read-only; append dated exports
Calibrated / filtered	data/02_processed/	Document transformations in src/

Figure-ready matrices	data/03_final/	Versioned; cite in publications
Methods and calibration	docs/	Linked from dataset metadata

Each dataset records who collected it, when, with which instrument, under which bias/temperature, and what processing was applied.

1.5 Expected data size

Category	Approximate size
Current repository	< 5 MB
Per measurement campaign	< 50 MB
Full project (estimated)	< 500 MB

1.6 Data utility

Target users: semiconductor device researchers, analog/RF circuit designers, reliability and yield engineers, reproducibility reviewers, and EOSC metadata harvesters.
Keywords: semiconductor characterization, CMOS, 180nm, IV curves, 1/f noise, threshold voltage, wafer yield, process corners, Monte Carlo, FAIR data.

2. FAIR data

2.1 Making data findable, including provisions for metadata

Measure	Status	Planned action
Persistent identifier (DOI)	Planned	Zenodo deposition (EOSC-aligned)
Version control	Active	Git tags per release (e.g. v1.0.
Descriptive metadata	Active	README + this DMP
Sidecar metadata	Planned	metadata/ YAML per DataCite Sche
Discovery indexing	Partial	GitHub; full after Zenodo DOI

Minimum metadata per deposition: title, creators, description, keywords, licence (CC-BY-4.0), version, publication date, related identifiers. Dublin Core and DataCite-compatible fields will be used. Metadata will be harvestable via Zenodo and GitHub.

2.2 Making data accessible

Data class	Access	Licence	Restriction
Measurement CSV/DAT/TXT	Open	CC-BY-4.0	None; no personal data
Lab notes	Open	CC-BY-4.0	Redact third-party names on requ
180 nm PDK / foundry models	Not distributed	N/A	IPR / contractual (foundry NDA)
Pre-publication embargo	None currently	?	Max. embargo if patents filed

2.2.1 Repository
Primary working repository: GitHub (https://github.com/uzzielperez/CNM).
Archival repository (planned): Zenodo ? trusted, metadata-ready, EOSC-federated general-purpose repository.
2.2.2 Data
Principle: *As open as possible, as closed as necessary* (Horizon Europe Programme Guide).
Access protocol: HTTPS git clone and direct download. No authentication for open datasets.
2.2.3 Metadata
Metadata openly available under CC0. Metadata includes access conditions and contact point even if datasets are later restricted.
Contact: uzzielperez25@gmail.com

2.3 Making data interoperable

Repository structure:
Interoperability: UTF-8 encoding; SI units in headers; consistent column naming (vgs, vds, ids, freq_hz, temp_c); CSV RFC 4180 compliance.
? data/01_raw/ ? factory and instrument outputs (read-only)
? data/02_processed/ ? calibrated tables
? data/03_final/ ? publication-ready matrices
? src/ ? analysis pipelines (Python, MATLAB, Origin)
? DataManagementPlan/ ? formal DMP (this document)
? docs/ ? calibration sheets and lab notes
? metadata/ ? DataCite sidecars (planned)

2.4 Increase data re-use

Output	Licence	Citation requirement
Datasets	CC-BY-4.0	DOI + version tag
Software (src/)	MIT	Repository + release tag
Documentation	CC-BY-4.0	Same as datasets

Reuse tools: Python 3.10+ (pandas), MATLAB R2021a+, Origin 2021+. See repository README Section 9 and src/ for pipelines.
Downstream users must cite the DOI, not redistribute PDK content, and document derivative processing.

3. Other research outputs

Output	Location	Preservation
Analysis scripts	src/	Git + Zenodo archive
Figures from 03_final/	Publications / Zenodo	Link to source CSV
SPICE netlists	src/ or docs/	Git; exclude NDA decks
Presentations	DataManagementPlan/ or Zenodo	Supplementary material

Software documentation follows Lee et al. (2014) Ten simple rules for documenting scientific software.

4. Allocation of resources

Activity	Responsible	Effort
Data curation	Data steward	~2 h/month
DMP / README updates	Data steward	Quarterly
Zenodo DOI registration	Data steward	~1 h per release
Backup verification	Data steward	Annually
Storage	GitHub + Zenodo free tier	?0 at current volume

5. Data security

Risk	Mitigation
Overwrite of raw data	data/01_raw/ append-only; Git hi
Data loss	GitHub remote + Zenodo snapshots
Unauthorised modification	Branch protection; immutable rel
Confidential leakage	No PDK/foundry files in reposito
Local workstation loss	Push after each curated batch

Standard institutional IT policies apply. No sensitive personal data stored (Section 6).

6. Ethics

Topic	Assessment
Human subjects	Not applicable ? device measurem
GDPR	Not applicable for current asset
Intellectual property	Measurement data CC-BY-4.0; PDK
Export control	Characterization results only; r
Ethics committee	Not required for ex vivo device

7. Other issues

End of Data Management Plan
? Legacy path datasets/ is being migrated to data/01_raw/, data/02_processed/, data/03_final/.
? Zenodo DOI and ORCID to be added before formal grant submission.
? Grant agreement number and institution to be inserted when CNM is linked to a funded Horizon Europe project.
? This DMP is a living document; review annually or before each Zenodo release.